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Grammar of the Wave

Towards Explainable Multivariate Time Series Event Detection via Neuro-Symbolic VLM Agents

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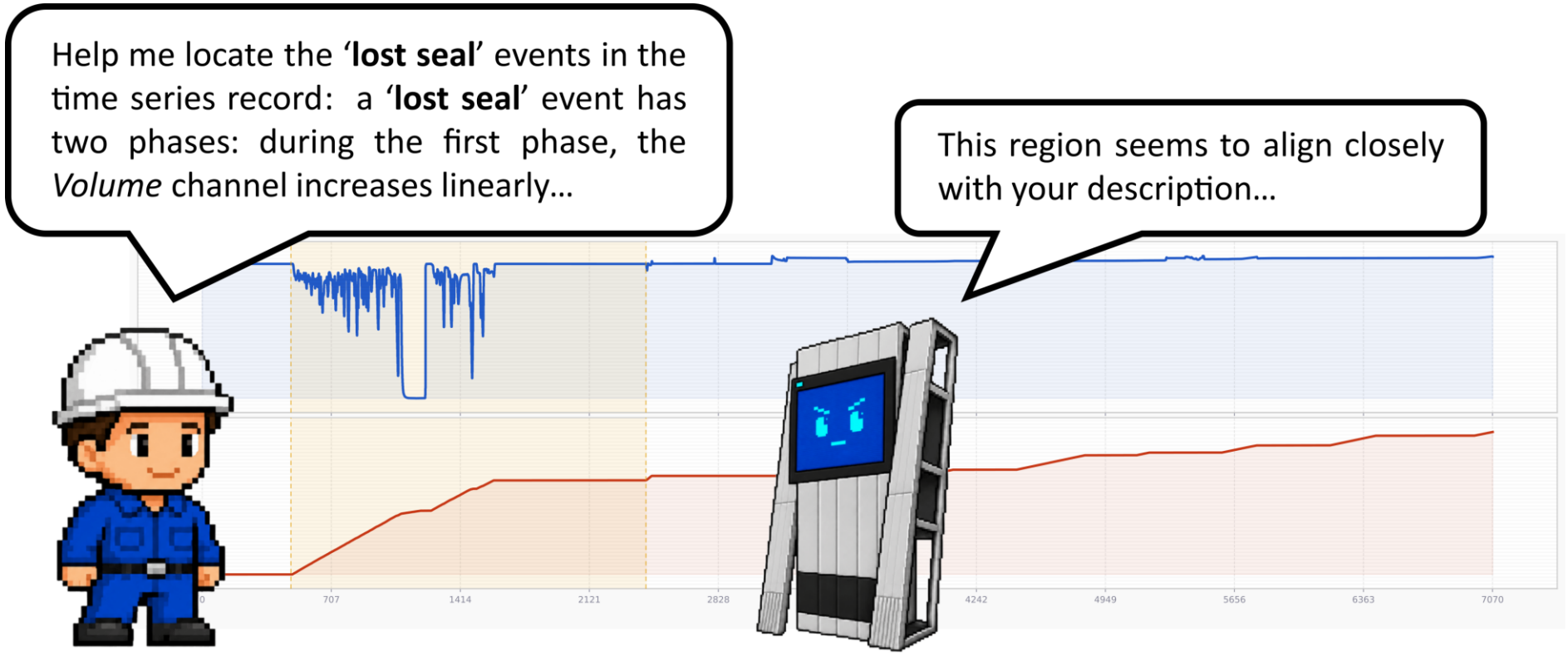
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Language-guided event detection

Expert descriptions are available. Dense event labels are expensive.



Input: signals + descriptions

Output: event intervals + classes

The grounding challenge

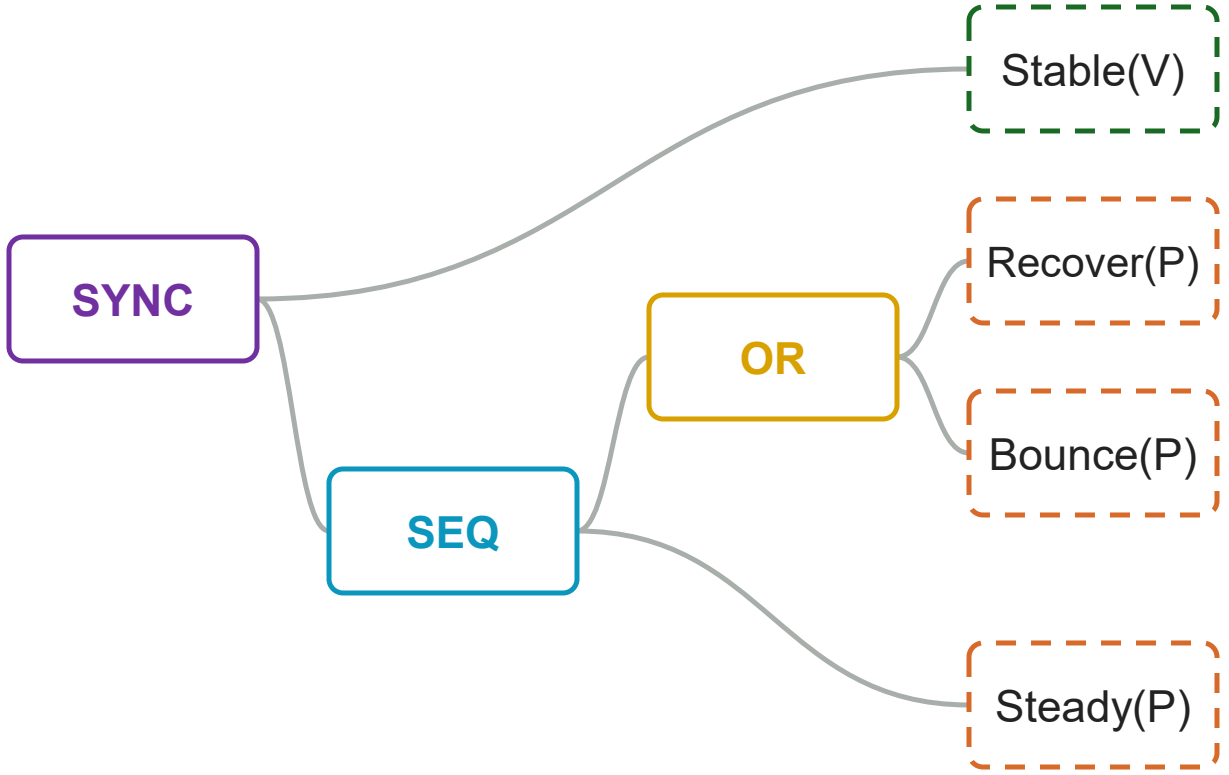
Semantic alignment, multichannel structure and verifiable evidence

Alignment	Map phrases such as “recover slightly” to signal morphology, with flexible timing and duration.
Multichannel	Combine patterns across channels through temporal relations, such as pressure recovery while volume stays stable.
Explainability	Localize the supporting primitives and expose their relations, so experts can check why an event was detected.

Event Logic Tree

Temporal operators build on Allen's interval algebra (1983)

> "During the **Buildup** event, the *Volume* should remain stable across the entire phase. Simultaneously, the *Pressure* should either recover slightly with the rate of change decreasing or bounce up quickly, and then reach a steady phase."



V: Volume P: Pressure



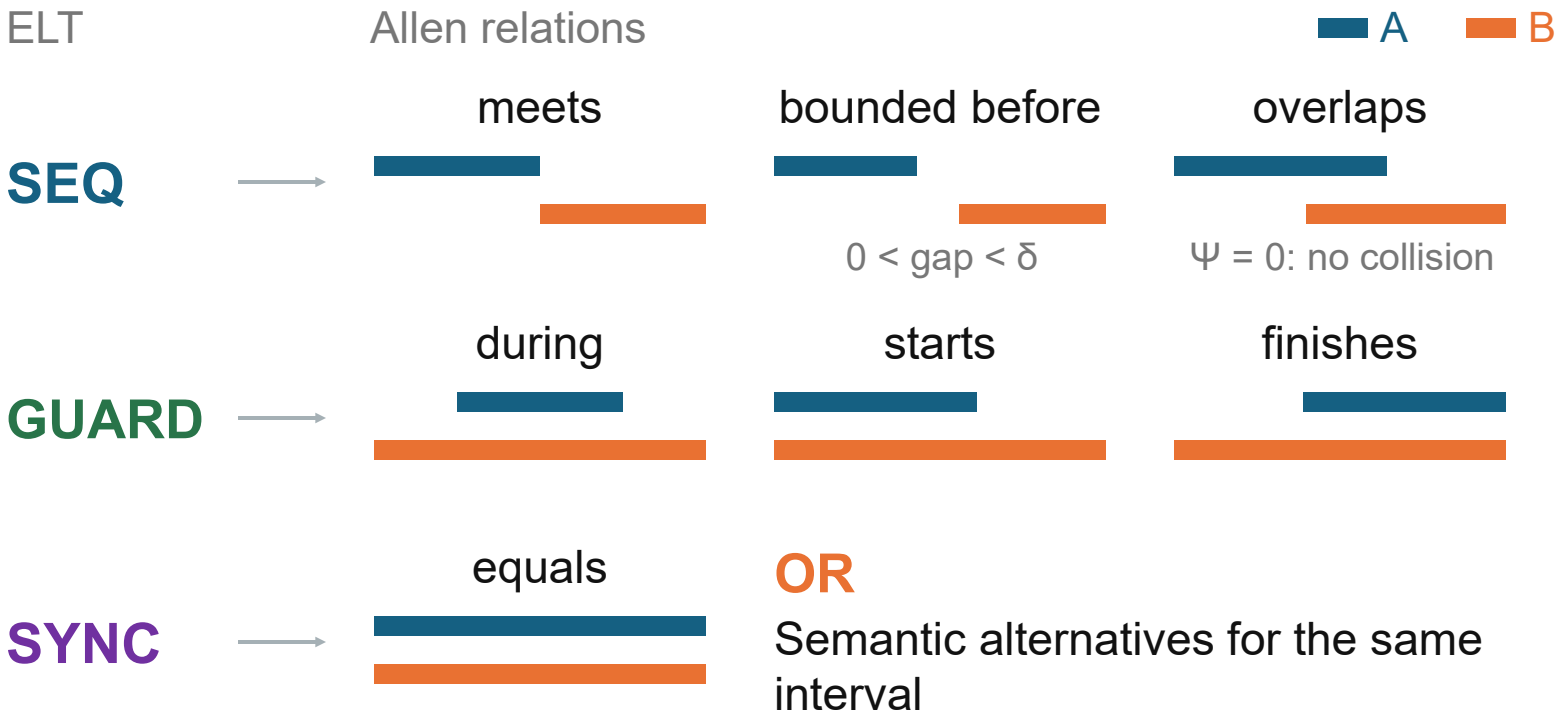
James F. Allen

13 basic interval relations (1983)

A classic theory of temporal reasoning

ELT operators and Allen relations

Allen's interval algebra, adapted to compact event grounding



Swapping A and B gives the six converse relations:

met-by, after, overlapped-by, contains, started-by, finished-by

Self-contained, compact events

The semantics must cover the event's full temporal span.



Exclude unrestricted disjoint before / after

Only a small gap < δ is tolerated as relaxed meets.

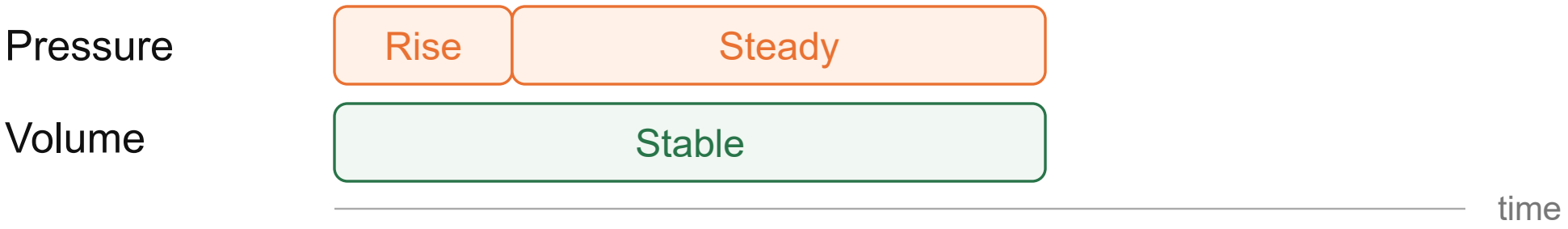
Allen (1983). ELT: Appendix A.2–A.4, especially Temporal Compactness and Proposition 1.

Topological elasticity

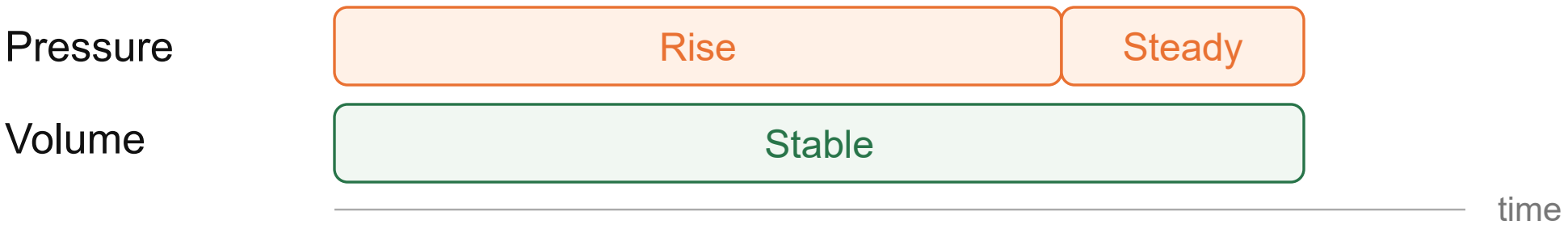
Durations can change while event logic stays valid

`SYNC(SEQ(Rise(P), Steady(P)), Stable(V))`

Instance A: short rise, long steady



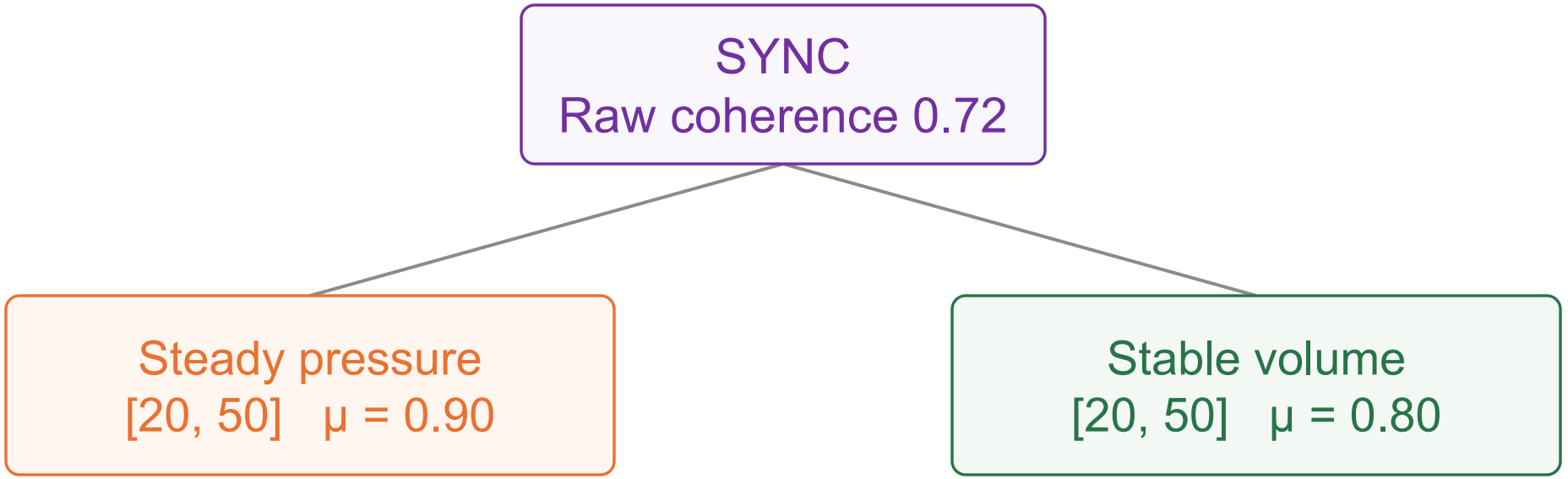
Instance B: long rise, short steady



Relative phase lengths vary. Order and synchronization remain.

Quantifiable evidence

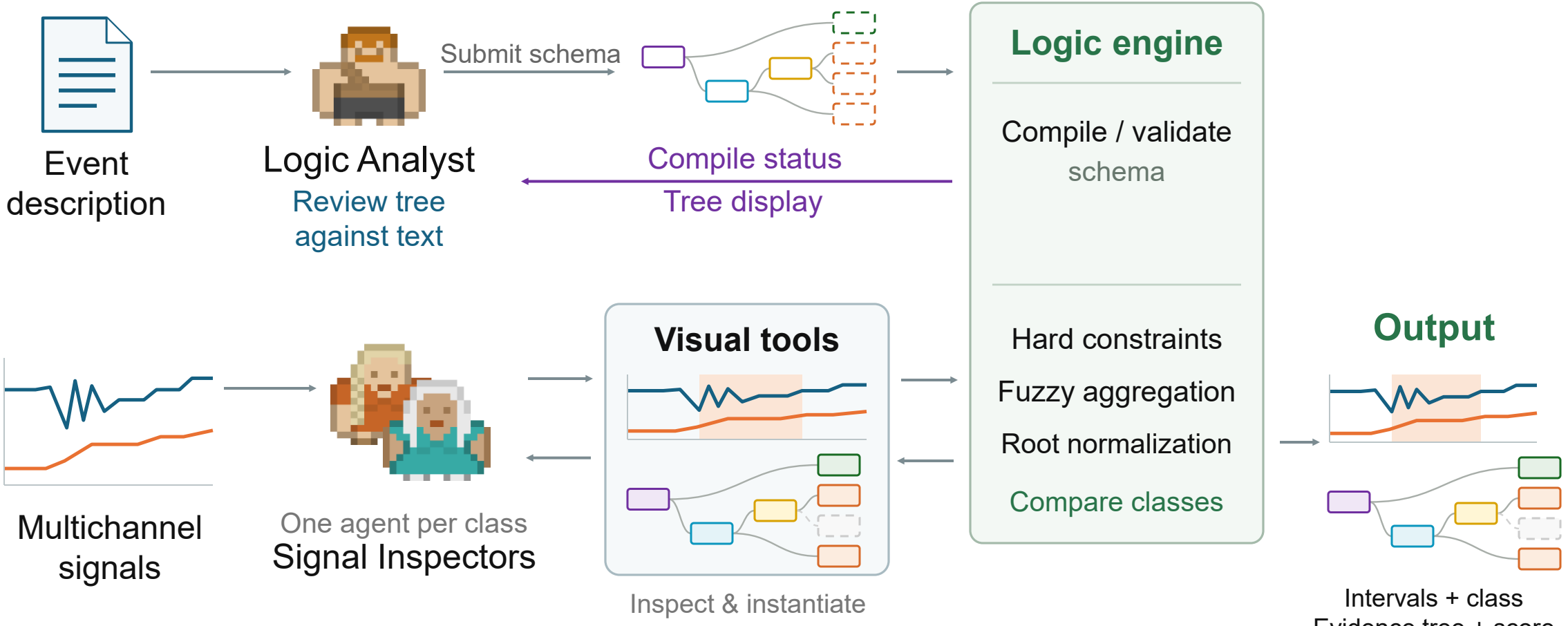
Primitive evidence combines under temporal constraints



Normalized root score: $\sqrt{0.72} \approx 0.85$

SELA: a multi-agent system

A reusable schema guides class-specific grounding and explains each detection



KITE: knowledge meets signals

Real events with expert descriptions and interval labels

Dataset	Samples	Channels	Event types
Pressure Test 41 North Sea jobs	62	2	Valid Test Lost Seal
Climate Hazard Open data NOAA + NWS	86	7	Dense Fog Frost Freeze Extreme Cold

Benchmarking ELT parsing

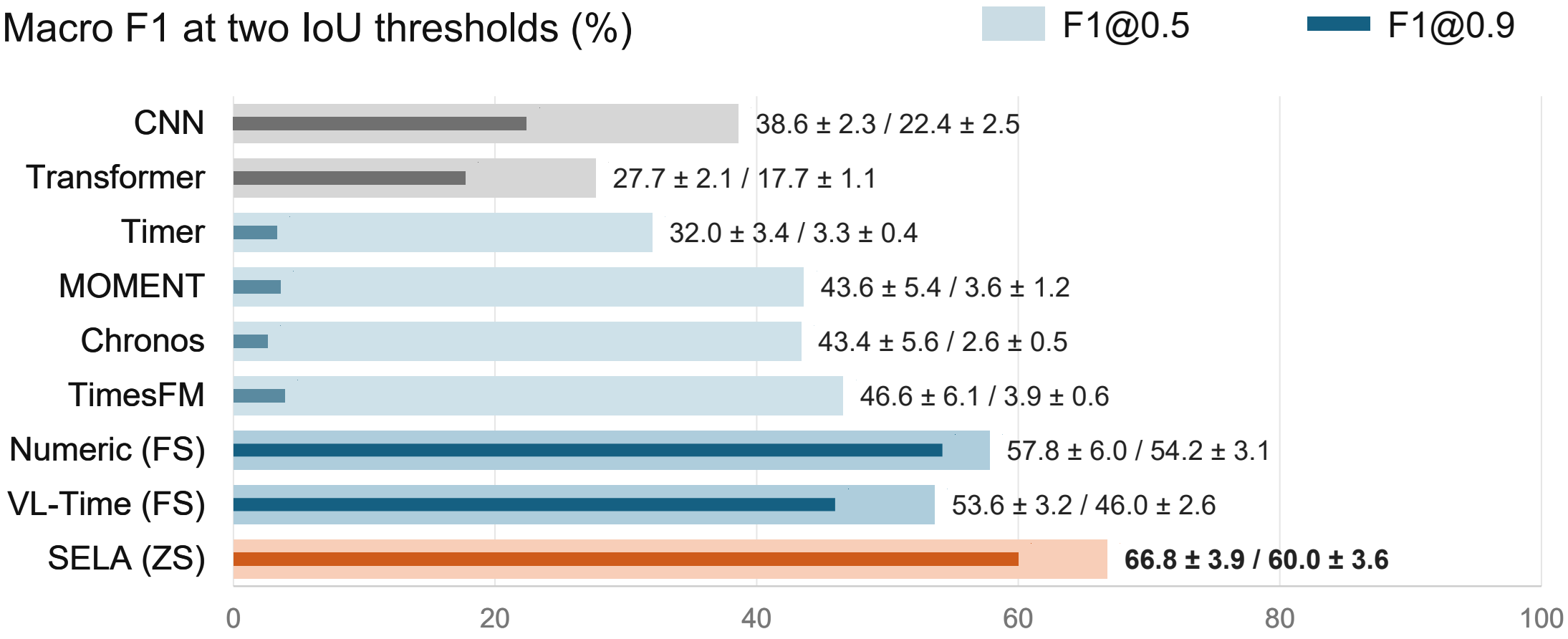
Can models recover event structure from language?

All valid parses reach 1.00 primitive accuracy.

Model	Pass	Structure	Accuracy	Coverage	Semantics
GPT-5	1.00	0.90	1.00	1.00	1.00
GPT-4.1	1.00	0.86	1.00	0.98	0.98
Qwen-3.5 27B	0.88	0.92	1.00	1.00	0.99
Gemma-4 31B	1.00	0.70	1.00	0.96	0.85

Qwen: strongest structure when valid, but the lowest pass rate.

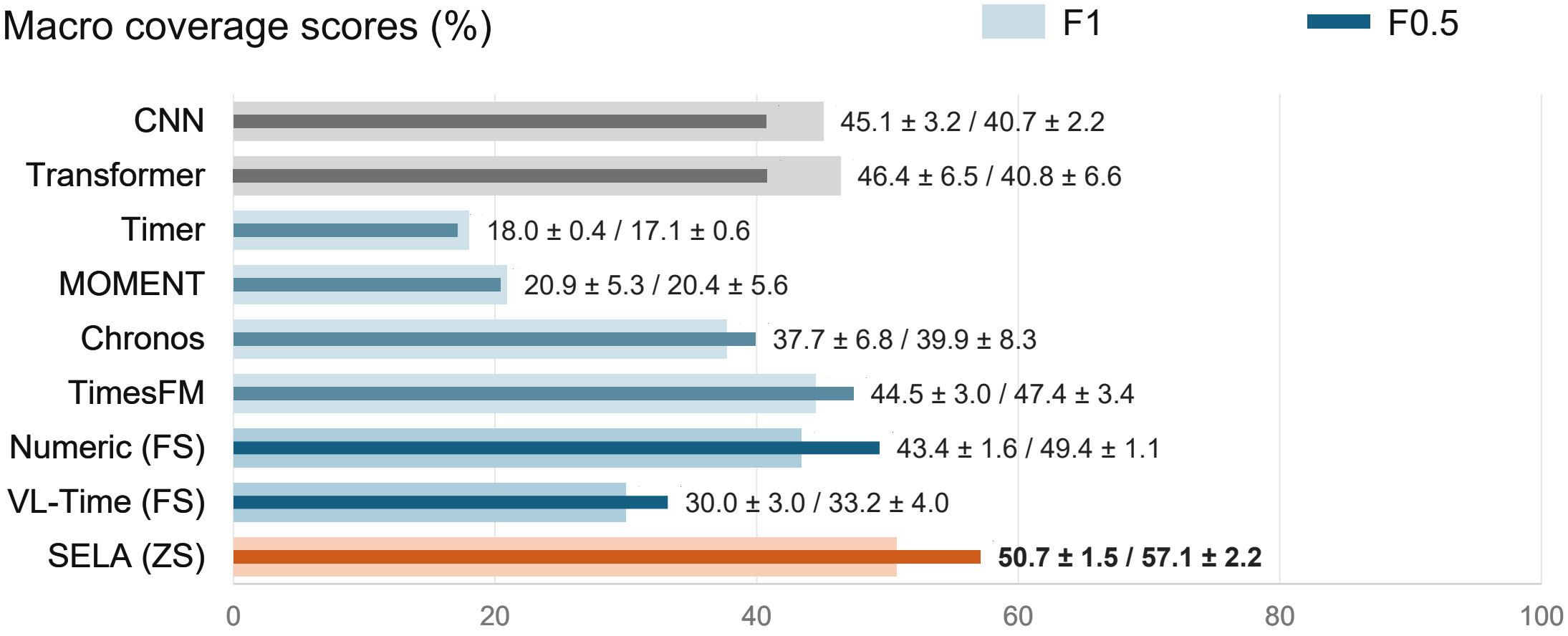
Pressure Test: event detection



CNN / Transformer | Time-series foundation models | LLM / VLM methods | **SELA**

Labels: F1@0.5 / F1@0.9. First six: supervised. GPT-5: FS = few-shot, ZS = zero-shot.

Climate Hazard: event detection

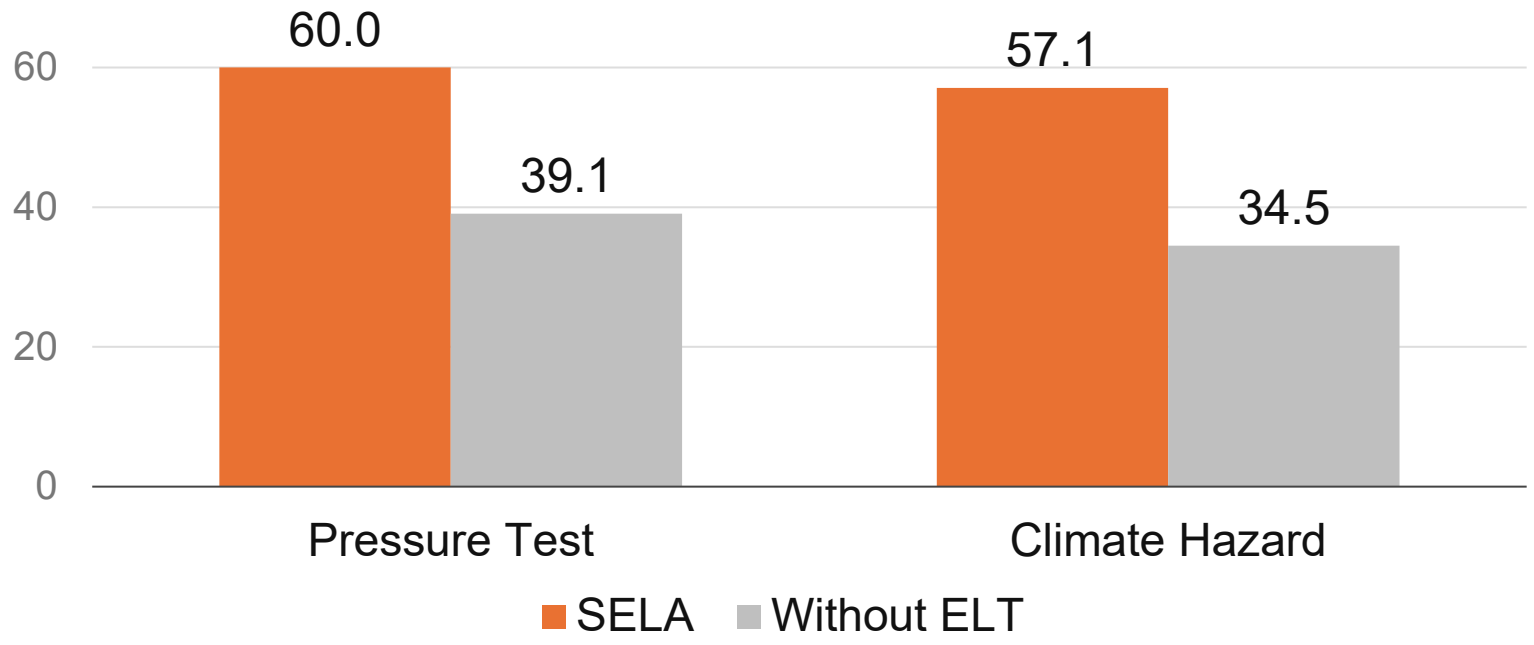


CNN / Transformer | Time-series foundation models | LLM / VLM methods | **SELA**

Labels: F1 / F0.5. First six: supervised. GPT-5: FS = few-shot, ZS = zero-shot.

Event logic drives the gains

Flat confidence aggregation loses the event structure



−20.9

points on
strict localization

−22.6

points on
Climate Hazard

GPT-5. Pressure Test: macro F1@0.9. Climate Hazard: macro coverage F0.5.

sela_inspector_valid_test

PT_46710318 / PressureTest / inspector trace replay

01 / 09

36 s loop · 2560 × 720

Inspector Reasoning

Survey the full sample

TRACE EXCERPT · reasoning #42

“First, I’ll survey the entire sample with view_full to understand the overall morphology on volume and pressure”

TOOL CALL · #043

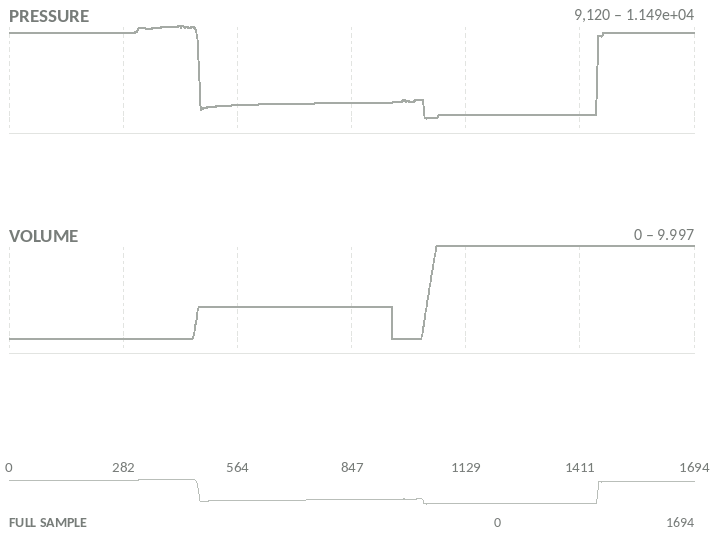
view_full()
pressure, volume · 1695 rows

Look for Pre, Drawdown, Buildup and Post before placing any primitive.

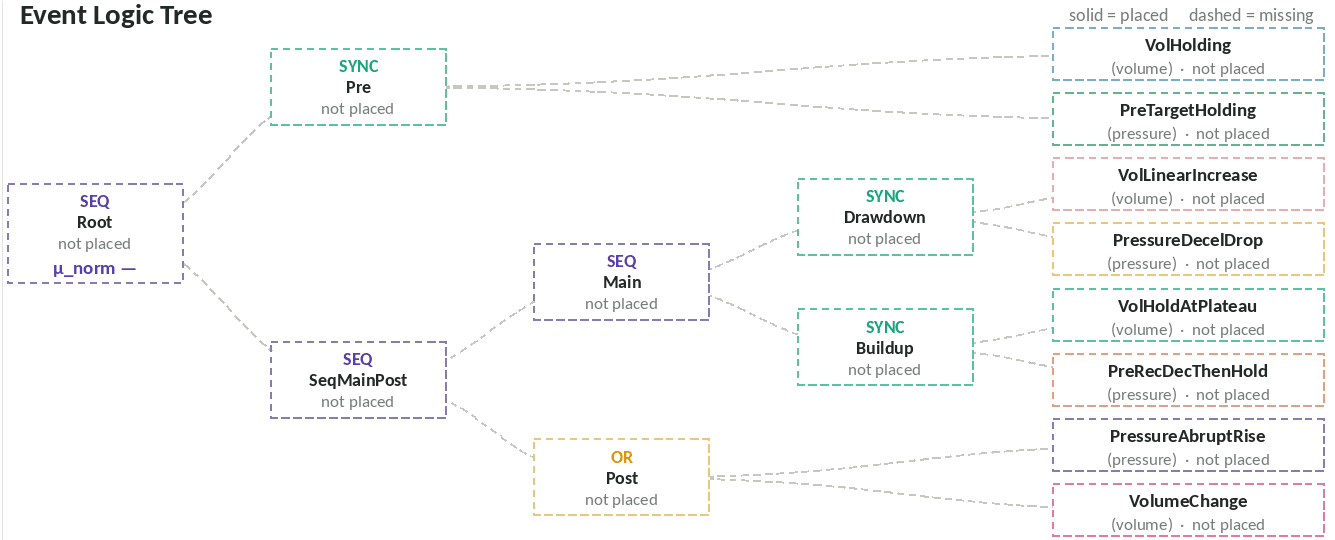
μ_norm awaiting placement

Signals & Primitive Instances

rows 0–1694



Event Logic Tree



Recorded trace · editorial pacing, not wall-clock time

Pixel characters: Kenney / CC0

sela_inspector_lost_seal

PT_46710318 / PressureTest / inspector trace replay

01 / 09

36 s loop · 2560 × 720

Inspector Reasoning

Survey the full sample

TRACE EXCERPT · reasoning #38

“I will first survey the entire sample to locate all regions that could plausibly match the lost seal phases, comparing volume/pressure behaviors and pre/post triggers.”

TOOL CALL · #039

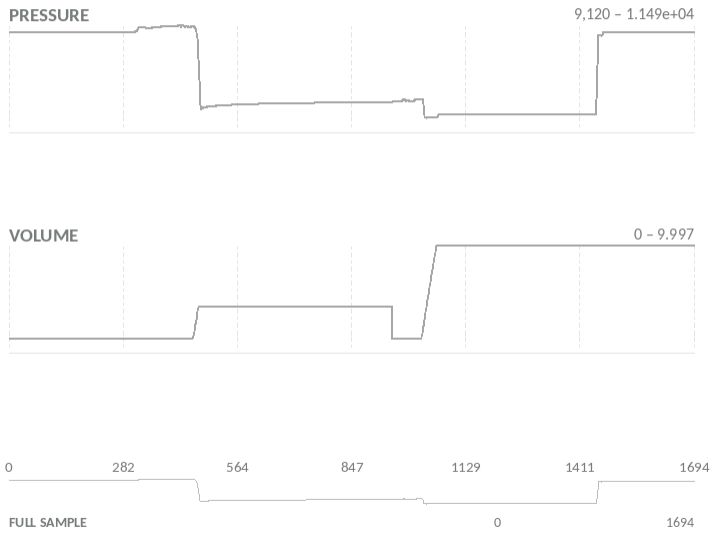
view_full()
pressure, volume · 1695 rows

Inspect both channels before placing any primitive.

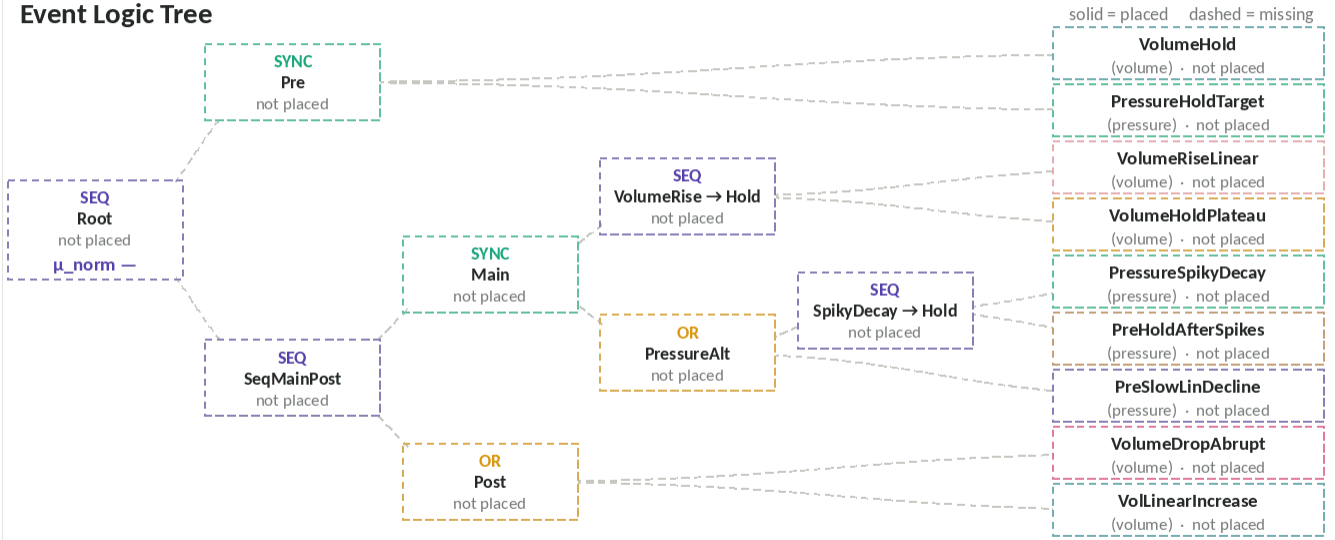
μ_norm awaiting placement

Signals & Primitive Instances

rows 0–1694



Event Logic Tree



Recorded trace · editorial pacing, not wall-clock time

Pixel characters: Kenney / CC0

Takeaways

Expert descriptions can guide zero-shot detection.

ELT turns domain knowledge into reusable constraints for signal grounding.

Temporal structure matters for precise localization.

Removing ELT costs 20.9 points on Pressure Test F1@0.9 (GPT-5).

Explanations should expose testable evidence.

ELT exposes primitive evidence
for expert review. Primitive-level alignment
still needs objective validation.

Paper

Code & Data